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Application Dossier
AI Benchmark Standards
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INTERACTIVE DEPLOYMENT URLS
Direct links to the live interactive web portfolio, deployed PDF, and source code:

🌐 Live Web Portfolio | 📄 Deployed PDF Dossier | 💻 GitHub Repository

18+	95% CI	0.94	p < 0.001
Frontier Models Audited	**Bootstrap Protocol**	**Inter-Rater Agreement**	**Significance Criterion**
MMLU-Pro, GSM8K, SWE-bench	B = 10,000 Resamples	Cohen's Kappa (κ) Standard	FDR / Bonferroni Corrected

EXECUTIVE PROFILE & ANALYTICAL PHILOSOPHY

Data Analyst and Subject Matter Expert specializing in AI model benchmarking, statistical chart analysis, error distribution auditing, and evaluation governance. Proven methodology in differentiating genuine architectural reasoning breakthroughs from stochastic evaluation noise, detecting pre-training test-set contamination (perplexity shift & n-gram memorization), resolving Simpson's Paradox in disaggregated benchmarks, and calibrating LLM-as-a-Judge systems to eliminate length and position biases.

MARKET-RELEVANT AI BENCHMARK PROJECTS

1. Multi-Model AI Benchmark Analytics Engine & Pareto Frontier

Statistical Frontier Modeling

Evaluates frontier models (Claude 3.5 Sonnet, GPT-4o, DeepSeek-R1, DeepSeek-V3, Llama 3.3 70B, Qwen 2.5 72B) across MMLU-Pro, GSM8K, SWE-bench Lite, and HumanEval, mapping multi-objective Pareto efficiency curves across Accuracy vs. Latency (ms) and Inference Cost ($/1M tokens).

- **Confidence Bounds:** Employs 10,000 non-parametric bootstrap iterations to compute exact 95% Confidence Intervals, preventing false claims of superiority when scores fall within the error band.
- **Tradeoff Discovery:** Proved that while DeepSeek-R1 leads in complex STEM reasoning (84.0% on MMLU-Pro), Claude 3.5 Sonnet (79.2%) is 2.7x faster (420ms vs 1,150ms), defining the optimal low-latency Pareto frontier for production SLAs.

Model	Family	MMLU-Pro	95% Confidence Band	Std Error (SE)	Latency (ms)	Pareto Efficient?
DeepSeek-R1	Reasoning-RL	**84.0%**	[82.6%, 85.3%]	±0.68%	1150 ms	**Yes ★**
Claude 3.5 Sonnet	Proprietary	**79.2%**	[77.8%, 80.5%]	±0.69%	420 ms	**Yes ★**
GPT-4o	Proprietary	**77.4%**	[75.9%, 78.8%]	±0.74%	380 ms	**Yes ★**
DeepSeek-V3	Open-Weights	**75.9%**	[74.4%, 77.3%]	±0.74%	310 ms	**Yes ★**
Llama 3.3 70B	Open-Weights	**71.8%**	[70.2%, 73.3%]	±0.79%	290 ms	**Yes ★**

2. Benchmark Anomaly & Contamination Forensic Suite

Data Integrity & Outlier Audit

Engineered an automated data integrity auditor detecting memorization vs. generalization through token perplexity shifts and multi-domain Simpson's Paradox audits.

- **Contamination Flagging ($Z > 2.5$):** Measured $\Delta \text{PPL} = \text{PPL}_{\text{unseen}} - \text{PPL}_{\text{test}}$. Discovered that fine-tuned open models claiming >90% GSM8K collapsed to 48.6% when numerical operands were perturbed, demonstrating test-set memorization rather than arithmetic generalization.
- **Simpson's Paradox Resolution:** Disaggregated macro benchmark scores into stratified sub-domains (Law, Medicine, Math, CS). Showed that Model Beta led macro scores (+1.4%) solely through high-volume CS syntax while suffering severe regressions (-12.4%) in Legal reasoning.

3. Bradley-Terry Elo Matrix & LLM-as-a-Judge Calibration

Tournament Modeling & Bias Mitigation

Constructed a head-to-head pairwise tournament model across 14,500 blind battle matches, resolving systematic evaluator distortions.

- **Length & Position Bias Elimination:** Identified uncalibrated judges granting an unmerited +7.3% win rate boost to verbose answers regardless of substance. Implemented a ridge regression length penalty, improving alignment with human annotators to $r = 0.91$ and Cohen's $\kappa = 0.89$.
- **Radar Capability Profiling:** Multi-dimensional mapping of models across Reasoning, Math, Coding, Instruction Following, Safety, and Hallucination Resistance.

4. Statistical Sample Size & Margin of Error Simulator

Interactive Pedagogical Laboratory

Designed an interactive mathematical laboratory enabling non-technical stakeholders and evaluation engineers to simulate sample sizes ($N=100$ to $5,000$), observed accuracy deltas, and task variances. Computes live margins of error, Z-scores, two-tailed p-values, and statistical power ($1 - \beta$).

APPLIED INDUSTRY CASE STUDIES

Case 01: Uncovering Simpson's Paradox in Multilingual Reasoning Benchmarks

Problem: A model release candidate claimed a +4.1% aggregate gain across 14 languages, but regional teams reported degraded low-resource customer sentiment handling.
Methodology: Disaggregated performance by language cohort. Discovered a 20x higher sampling density in English masked an 8.4% drop in Indic and African languages.
Outcome: Implemented stratified invariant population weighting, halting a flawed release and establishing a new corporate multilingual benchmarking standard.

Case 02: Forensic Audit of Math Test-Set Leakage in 70B Parameter Models

Problem: An open-weights 70B model claimed higher GSM8K performance than proprietary models 5x its size.
Methodology: Executed an n-gram prefix memorization audit and compared token loss on original questions versus dynamically perturbed arithmetic variants.
Outcome: Proved performance collapsed from 91.2% to 48.3% on perturbed operands, confirming rote test-set contamination. Produced an executive forensic audit for safety leadership.

Case 03: Eliminating Position & Length Biases in Automated Evaluation

Problem: Automated LLM judges exhibited 64% first-token preference (order bias) and favored wordy responses.
Methodology: Formulated an Order A/B permutation swap protocol and fitted an orthogonal length de-biasing regression penalty.
Outcome: Elevated inter-annotator agreement from Cohen's $\kappa = 0.62$ to $\kappa = 0.89$, cutting human evaluation audit costs by 70%.

METHODOLOGICAL COMPETENCY & TECHNICAL STACK

Statistical Methods

- Non-parametric Bootstrapping (10k)
- Paired Wilcoxon & Welch's t-test
- Multiple Testing (Bonferroni / FDR)
- Bradley-Terry Tournament Modeling
- Cohen's Kappa (κ) & Gwet's AC1

AI Evaluation Frameworks

- MMLU-Pro, GSM8K, MATH
- SWE-bench (Verified & Lite), HumanEval
- HELM, MT-Bench, LMSYS Arena
- TruthfulQA & Hallucination Auditing
- Inference Latency (TTFT) & TPS Analysis

Data & Engineering Stack

- Python (SciPy, Statsmodels, Pandas, NumPy)
- SQL, PostgreSQL, BigQuery
- Chart.js, D3.js, Matplotlib, Seaborn
- Git, CI/CD, Vercel, PyTest
- Executive & Technical Communication